



EAST TEXAS A&M
— UNIVERSITY —

CSCI 352.61E

DIGITAL FORENSICS

System Forensics, Investigation, and Response

Lesson 12

Performing Network Analysis

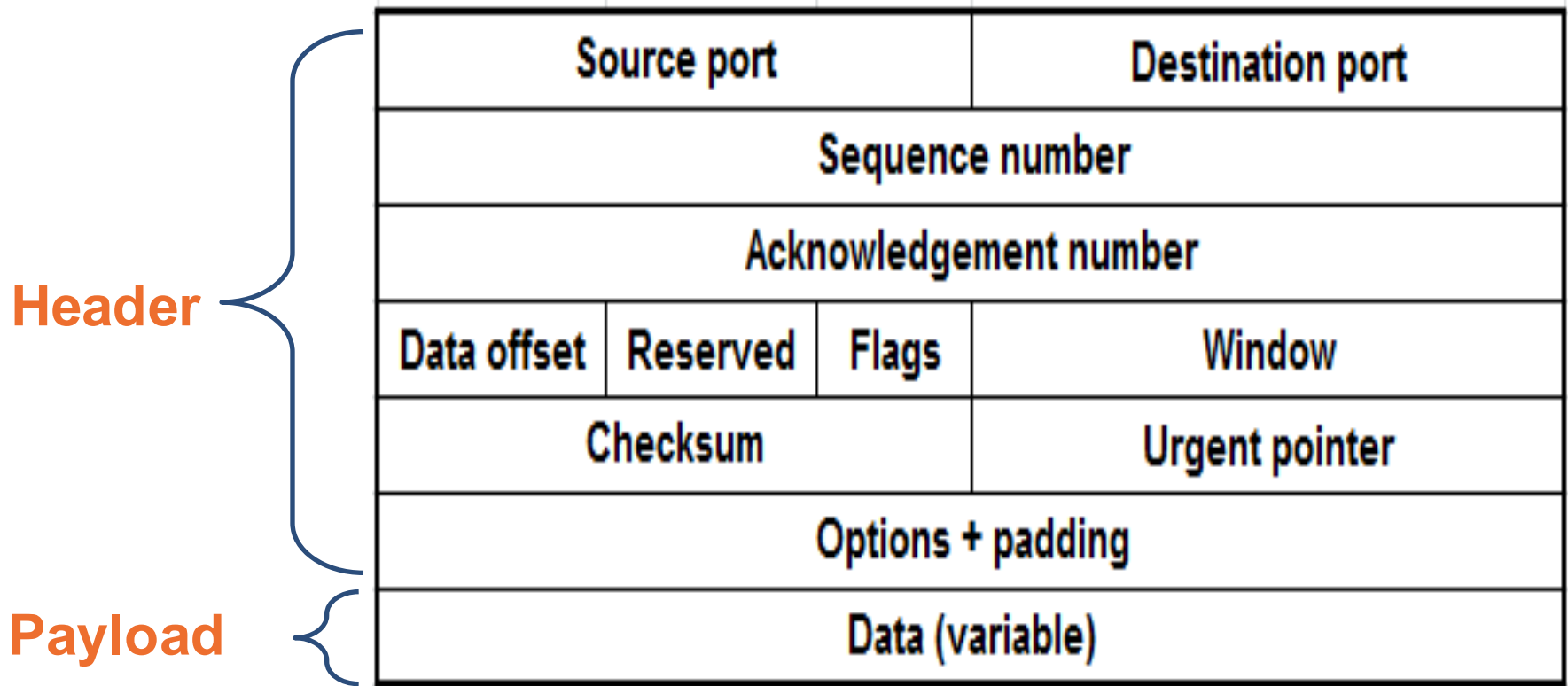
Learning Objective

- Explain how to perform a network analysis.

Key Concepts

- Network packet analysis basics
- Traffic analysis
- Router forensics
- Firewall forensics

Network Packets: Structure



Packet Header

Ethernet Header

Source
MAC

Destin-
ation
MAC

IP Header

IP
addresses

Protocol
#s

TCP Header

Ports

Seq. #

Data

TCP Header Bits, of Interest

URG

- Traffic marked as urgent
- Bit is rarely used

ACK

- Acknowledges the attempt to synchronize communications

RST

- Resets the connection

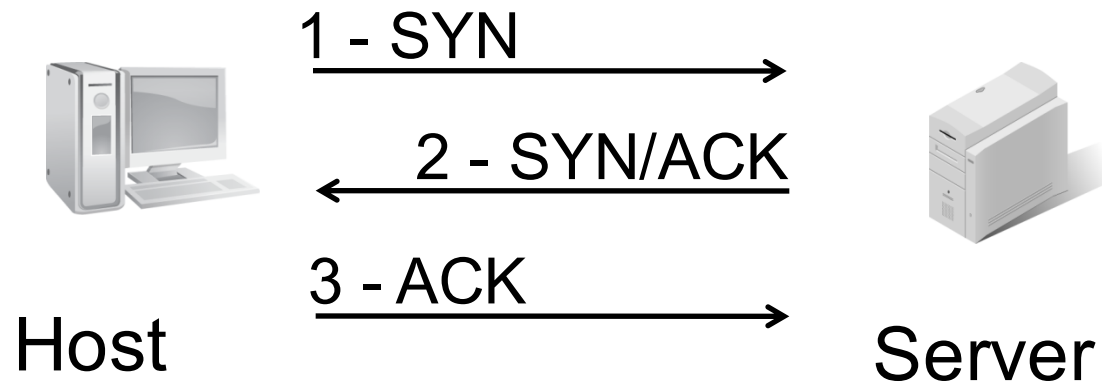
SYN

- Synchronizes sequence numbers

FIN

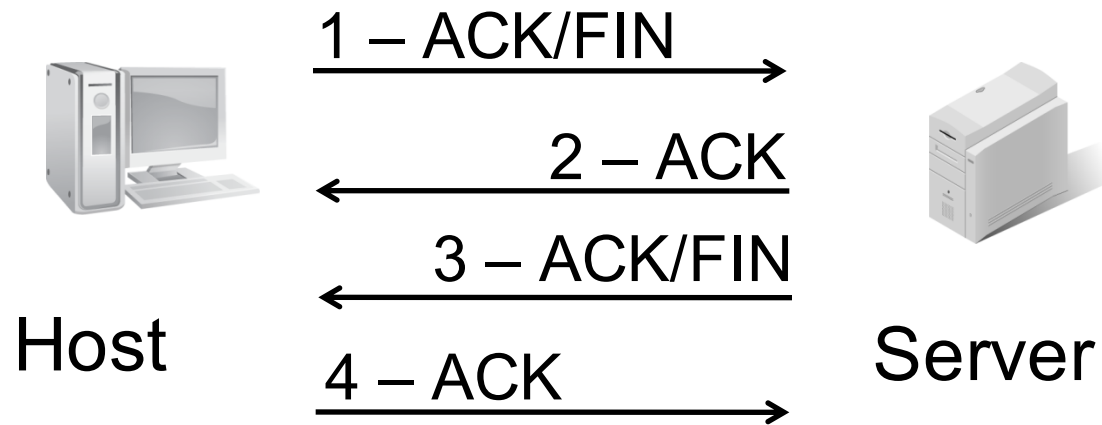
- Indicates there is no more data from the sender

TCP Three-Way Handshake



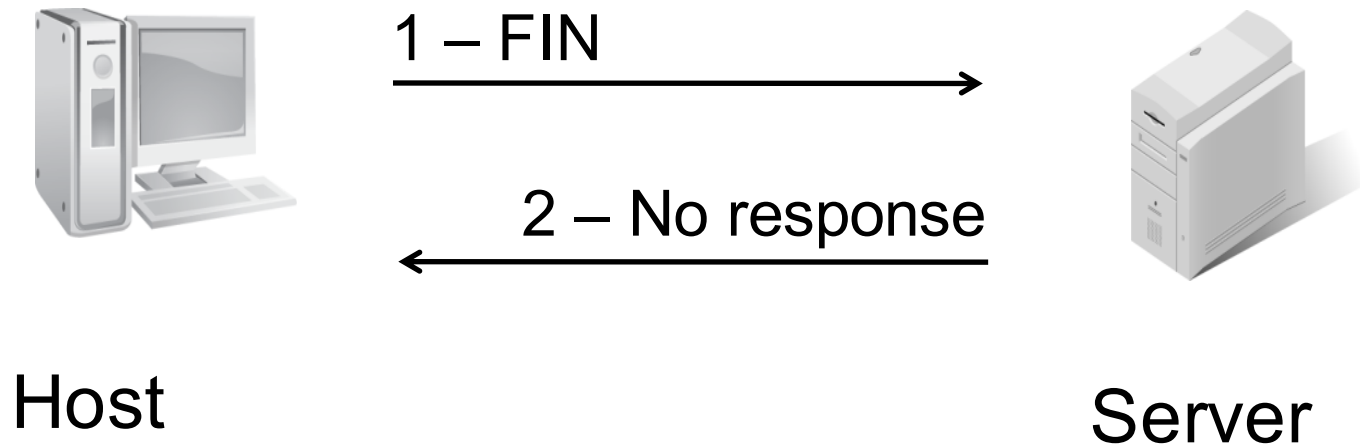
Synchronize (SYN)
Acknowledge (ACK)

TCP Connection Termination



Acknowledge (ACK)
Finish (FIN)

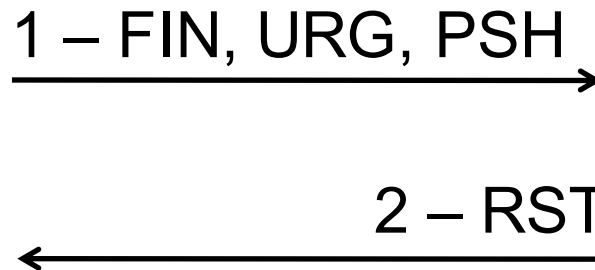
FIN Scan



Christmas Tree Scan



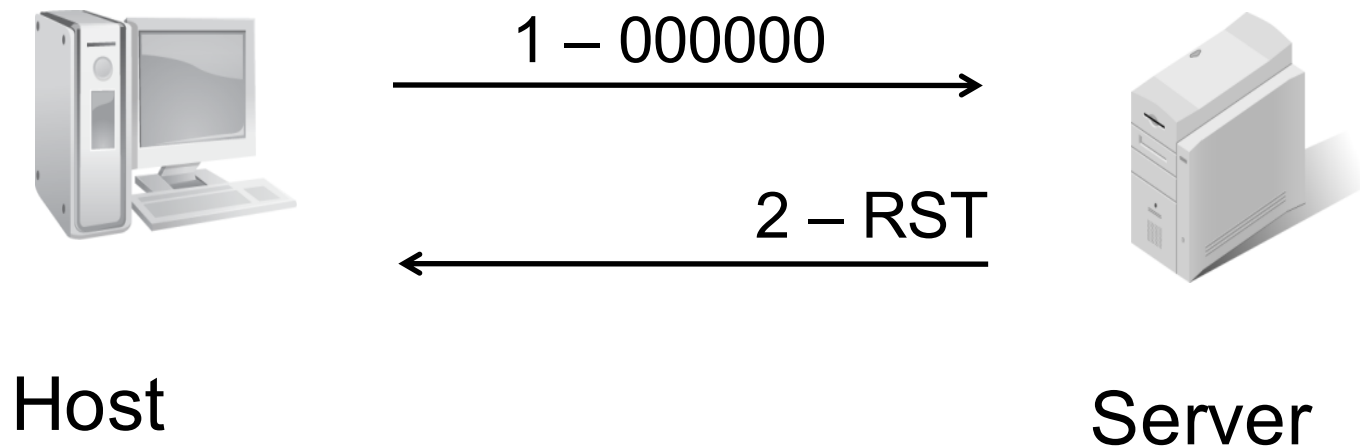
Host



Server

Finish (FIN)
Urgent (URG)
Push (PSH)
Reset (RST)

Null Scan



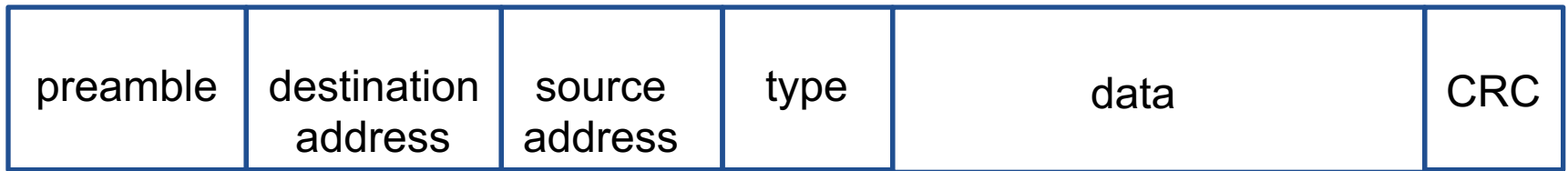
Network Packet: Payload

- The body or information content of a packet
- Actual content that the packet is delivering to the destination
- If packet is fixed length, payload may be padded with blank information or a specific pattern to make it the right size

Network Packet: Trailer

- Contains error-checking data to detect errors that occur during transmission
- May be part of the Ethernet or Point-to-Point Protocol (PPP) frame or other Layer 2 protocol

Cyclical Redundancy Check (CRC)



Why is the CRC almost always in a trailer, rather than in a header?

If CRC found, frame discarded

Ports

- A number that identifies a channel in which communication can occur
- 65,635 possible ports
- Knowing what port a packet was destined for (or coming from) tells you what protocol it was using

Some Important Ports

Port Number	Protocol that Uses Port
20, 21	File Transfer Protocol (FTP)
22	Secure Shell (SSH)
23	Telnet
25	Simple Mail Transfer Protocol (SMTP)
53	Domain Name Service (DNS)
80	Hypertext Markup Language (HTML)
110	Post Office Protocol Version 3 (POP3)

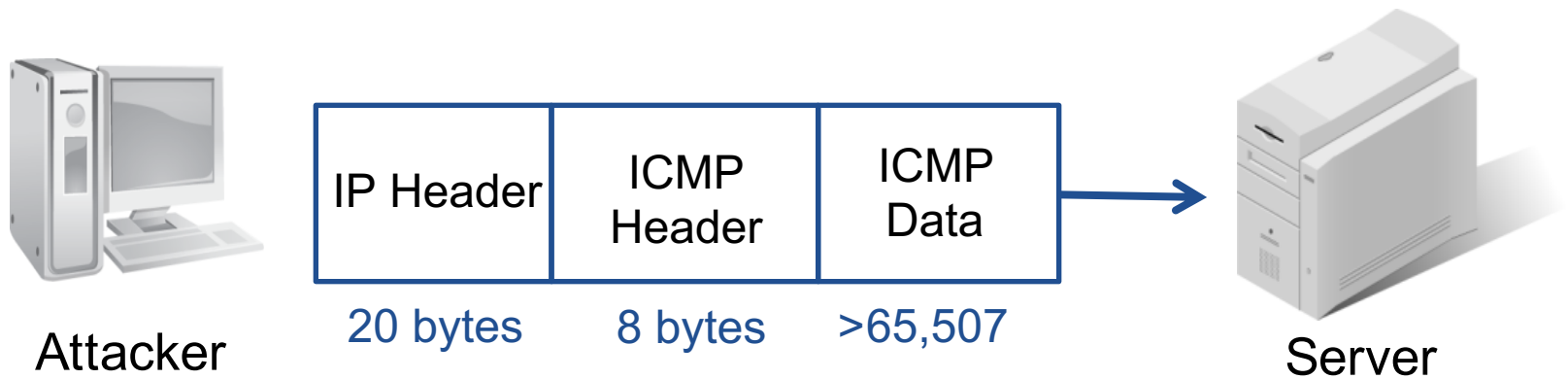
Important Intruder Ports

Port Number	Utility
407	Timbuktu
6666	Beast
23476/23477	Donald Dick
31337	Back Orifice
43188	Reachout
54320/54321	B02K

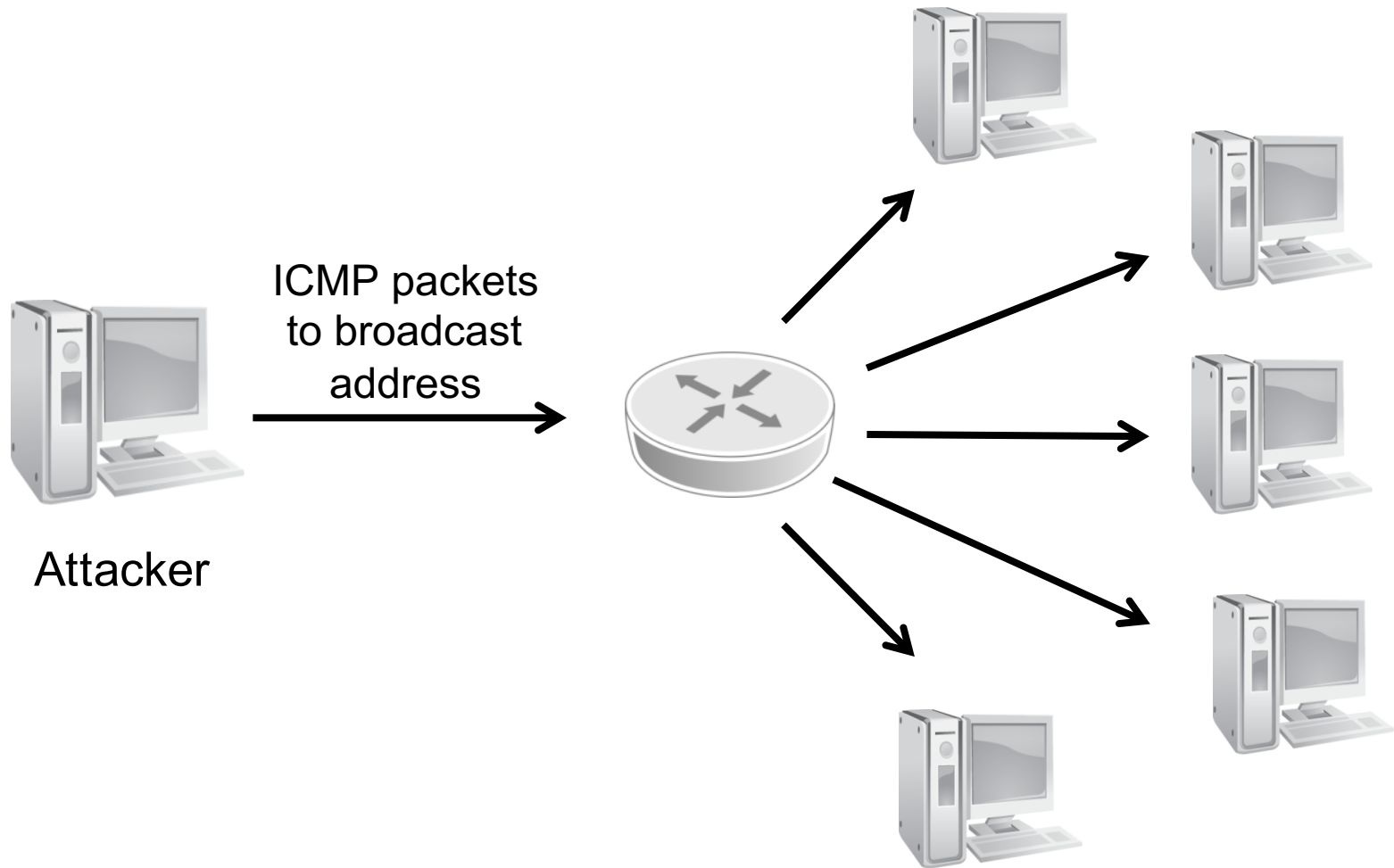
Denial of Service (DoS) Attacks

- Can be targeted at a given server; increased traffic affects rest of the target network
- Attacker usually floods the network with malicious packets, preventing legitimate network traffic from passing
- Attacker uses one of three approaches:
 - Damages the target machine's ability to operate
 - Overflows the target machine with open connections at the same time
 - Uses the bandwidth to the target machine

Ping of Death Attack



Ping Flood



Teardrop Attack

- Attacker sends fragments of packets with bad values in them
- Causes the target system to crash when it tries to reassemble fragments

SYN Flood Attack



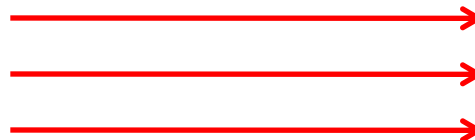
Attacker

1 – SYN →

← 2 – SYN/ACK

3 – SYN →

← 4 – SYN/ACK



...

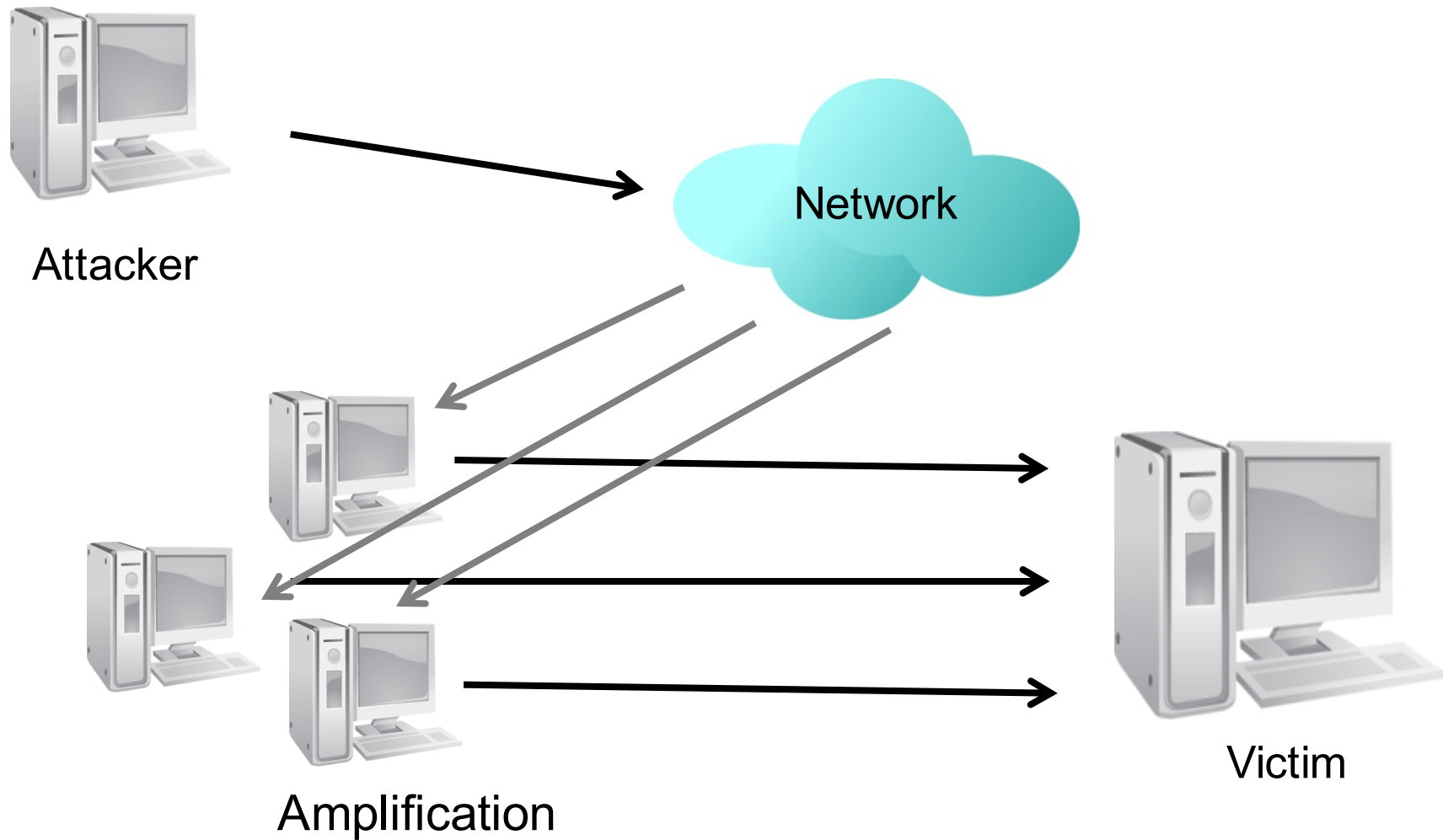


Server

Land Attack

- Attacker sends a fake TCP SYN packet with the same source and destination IP addresses and ports as the target computer
- Victim computer is tricked into thinking it is sending messages to itself because the packets coming from the outside use the computer's own IP address

Smurf Attack



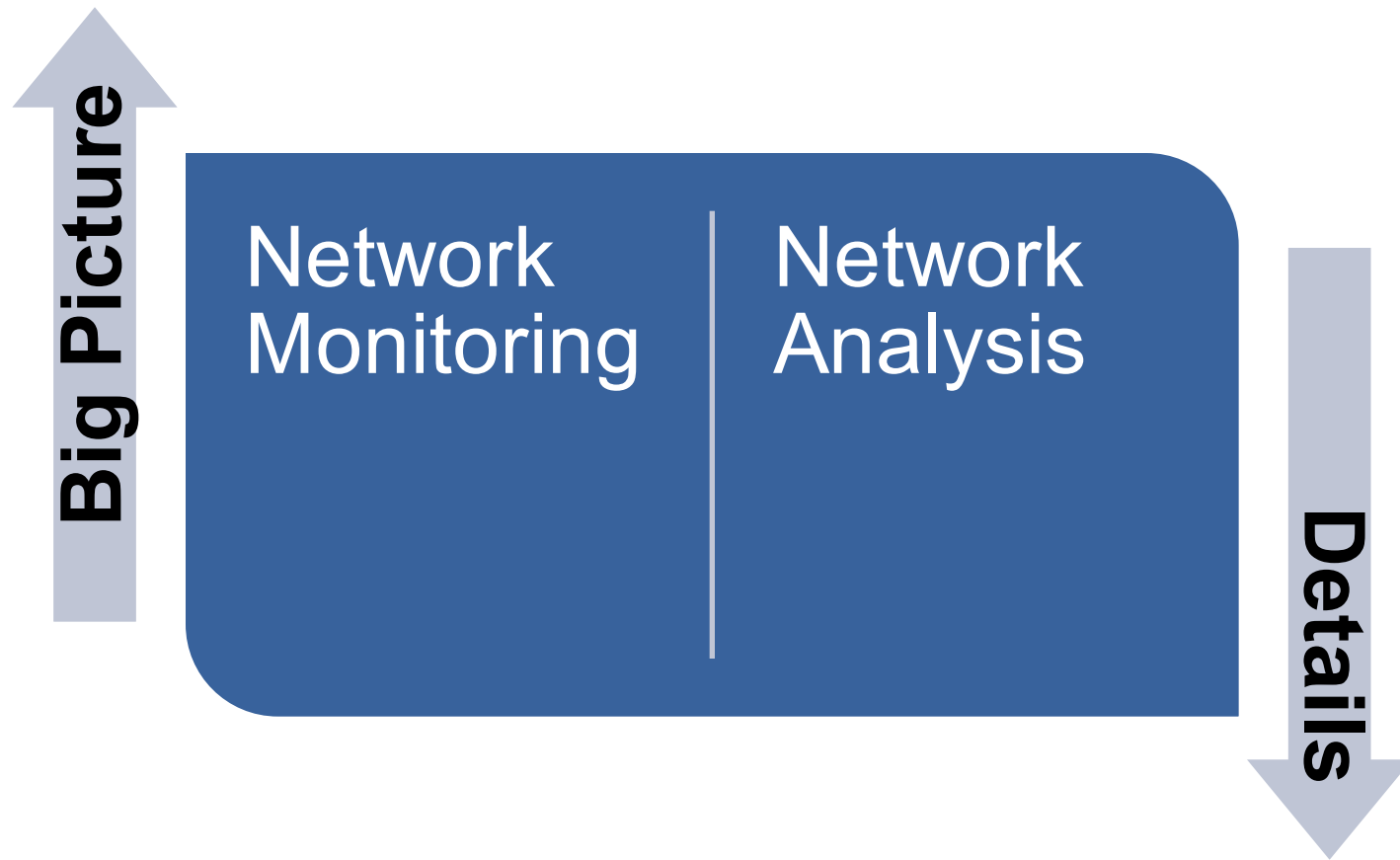
Fraggle Attack

- Similar to a smurf attack
- Fraggle uses spoofed User Datagram Protocol (UDP) packets instead of Internet Control Message Protocol (ICMP) echo replies
- Can often bypass a firewall

Packet Mistreating

- Occurs when a compromised router mishandles packets
- Results in congestion in a part of the network

Network Traffic Analysis Overview



What Is a Sniffer?

- Software or hardware that can intercept and log traffic passing over a digital network
- Extracts network packets and performs a statistical analysis on the dumped information
- Commonly applied sniffers include Tcpdump (UNIX platforms and) WinDump

Tcpdump and WinDump

- Tcpdump
 - A commonly applied sniffer for various UNIX platforms
- WinDump
 - A version of Tcpdump for Windows
- Both programs extract network packets and perform statistical analysis on dumped information

Wireshark

- Network protocol analyzer
- Captures Ethernet, IEEE 802.11, PPP/HDLC, ATM, Bluetooth, USB, Token Ring, Frame Relay, FDDI, and other packets
- Analyzes real-time and saved data
- Runs on Windows, Linux, OS X, Solaris, FreeBSD, NetBSD, and others
- Supports IPv4 and IPv6
- Allows Voice over IP (VoIP) analysis
- Freely available

Wireshark Packet Capture

The image shows a Wireshark packet capture window. The title bar indicates it is capturing from a Microsoft NPFF interface. The menu bar includes File, Edit, View, Go, Capture, Analyze, Statistics, Telephony, Tools, Internals, and Help. The toolbar contains various icons for file operations, capture control, and analysis. The packet list pane shows a table of captured packets with columns for No., Time, Source, Destination, Protocol, Length, and Info. The details pane for the selected packet (No. 40) shows the frame structure: Ethernet II, Internet Protocol Version 6, User Datagram Protocol, and Link-local Multicast Name Resolution (query). The packet bytes pane at the bottom displays the raw data in hexadecimal and ASCII.

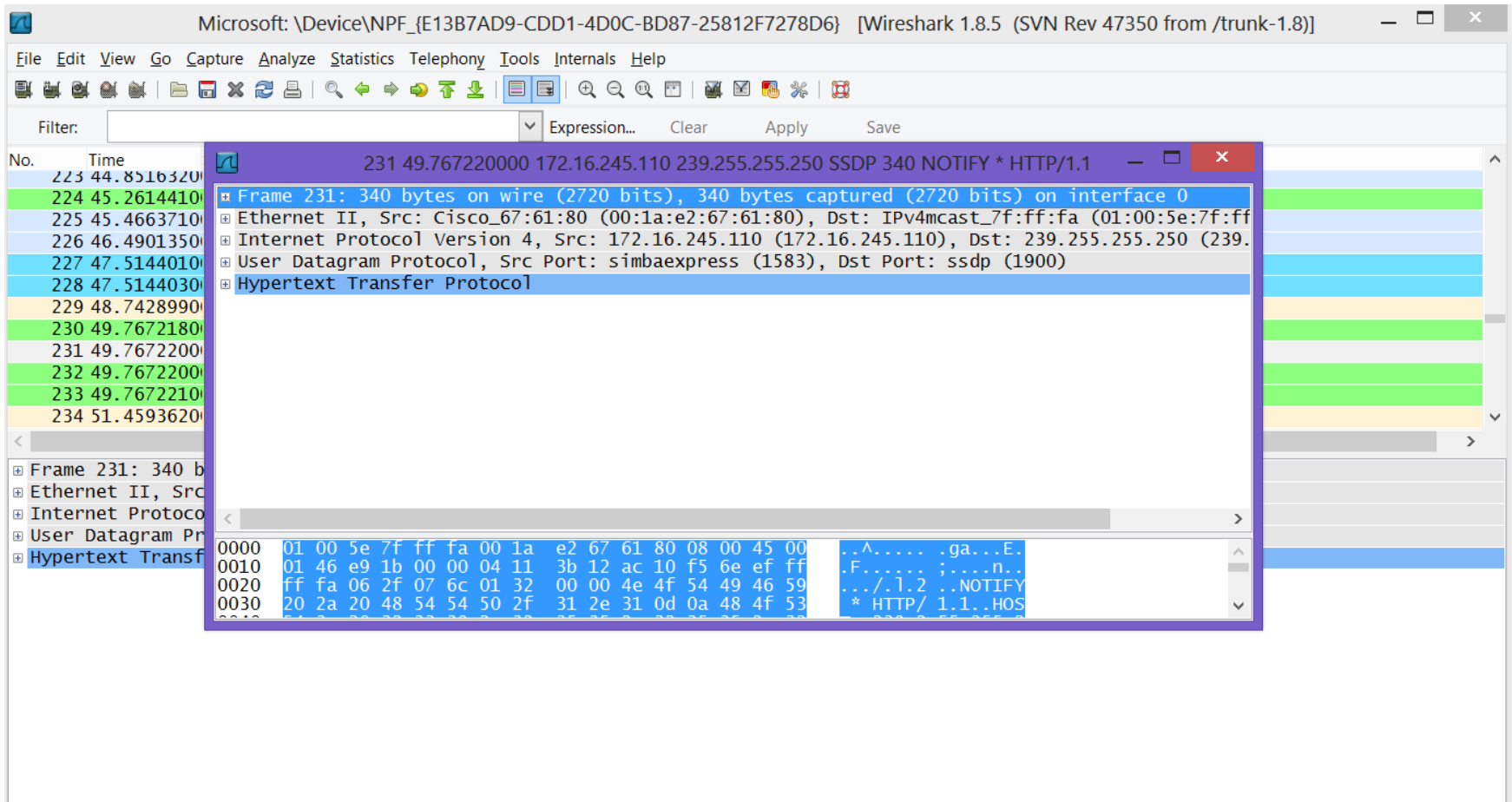
No.	Time	Source	Destination	Protocol	Length	Info
70	9.66641400	172.16.215.225	69.22.148.90	TCP	54	4642 > http [ACK] Seq=2 Ack=2 Win=1024 Len=0
71	9.66646800	172.16.215.225	69.22.148.90	TCP	54	4647 > http [ACK] Seq=2 Ack=2 Win=1024 Len=0
72	9.66652200	172.16.215.225	69.22.148.90	TCP	54	4644 > http [ACK] Seq=2 Ack=2 Win=1024 Len=0
73	9.66664700	69.22.148.90	172.16.215.225	TCP	60	http > 4643 [FIN, ACK] Seq=1 Ack=2 Win=8372 Len=0
74	9.66666800	172.16.215.225	69.22.148.90	TCP	54	4643 > http [ACK] Seq=2 Ack=2 Win=1021 Len=0
75	9.68495600	74.125.141.95	172.16.215.225	TCP	60	http > 4641 [FIN, ACK] Seq=1 Ack=2 Win=1000 Len=0
76	9.68503600	172.16.215.225	74.125.141.95	TCP	54	4641 > http [ACK] Seq=2 Ack=2 Win=1024 Len=0
77	9.70246000	172.16.245.111	239.255.255.250	SSDP	340	NOTIFY * HTTP/1.1
78	9.70246100	172.16.245.111	239.255.255.250	SSDP	340	NOTIFY * HTTP/1.1
79	9.70246200	172.16.245.111	239.255.255.250	SSDP	340	NOTIFY * HTTP/1.1
80	9.70246200	172.16.245.111	239.255.255.250	SSDP	340	NOTIFY * HTTP/1.1
81	9.97015100	172.16.215.225	69.4.231.52	TCP	54	4652 > http [FIN, ACK] Seq=1 Ack=1 Win=1024 Len=0

Frame 40: 95 bytes on wire (760 bits), 95 bytes captured (760 bits) on interface 0
Ethernet II, Src: LiteonTe_57:e2:aa (20:16:d8:57:e2:aa), Dst: IPv6mcast_00:01:00:03 (33:33:00:01:00:03)
Internet Protocol Version 6, Src: fe80::1171:716f:b974:9bd3 (fe80::1171:716f:b974:9bd3), Dst: ff02::1:3 (ff02::1:3)
User Datagram Protocol, Src Port: 51876 (51876), Dst Port: 11mnr (5355)
Link-local Multicast Name Resolution (query)

0000 33 33 00 01 00 03 20 16 d8 57 e2 aa 86 dd 60 00 33W.....
0010 00 00 00 29 11 01 fe 80 00 00 00 00 00 00 11 71 ...).....q
0020 71 6f b9 74 9b d3 ff 02 00 00 00 00 00 00 00 00 qo.t.....h...
0030 00 00 00 01 00 03 ca a4 14 eb 00 29 68 f2 17 8ah.....

Courtesy of the Wireshark Foundation

Wireshark Packet Details



Courtesy of the Wireshark Foundation

HTTPSniffer

HttpDetect (EffeTech HTTP Sniffer)

File View Sniffer Help

No.	Time	Client (IP:PORT)	Server (IP:PORT)	URL	File...	Status
69	Ju...	192.168.1.3 :3057	www.effetechn.com :80	/	35660	FIN, 200
70	Ju...	192.168.1.3 :3058	www.effetechn.com :80	/images/style.css		FIN, 304
71	Ju...	192.168.1.3 :3058	www.effetechn.com :80	/images/logo_main.jpg		FIN, 304
72	Ju...	192.168.1.3 :3058	www.effetechn.com :80	/images/chinese_edition.gif		FIN, 304
73	Ju...	192.168.1.3 :3058	www.effetechn.com :80	/images/space.gif		FIN, 304
74	Ju...	192.168.1.3 :3059	www.effetechn.com :80	/images/arrow_small.gif		FIN, 304
75	Ju...	192.168.1.3 :3058	www.effetechn.com :80	/images/award_tucows_4ratel...		FIN, 304
76	Ju...	192.168.1.3 :3060	www.etherdetect.com...	/images/logo_ms.gif	628	FIN, 200
77	Ju...	192.168.1.3 :3061	www.etherdetect.com...	/images/logo_ibm.gif	1217	FIN, 200
78	Ju...	192.168.1.3 :3059	www.effetechn.com :80	/images/award_FileHungry_5s...		FIN, 304
79	Ju...	192.168.1.3 :3058	www.effetechn.com :80	/images/award_softwareseeke...		FIN, 304
80	Ju...	192.168.1.3 :3061	www.etherdetect.com...	/images/logo_mit.gif	259	FIN, 200
81	Ju...	192.168.1.3 :3060	www.etherdetect.com...	/images/logo_ms.gif		Requested
82	Ju...	192.168.1.3 :3059	www.effetechn.com :80	/images/award_webaward2002e...		FIN, 304
83	Ju...	192.168.1.3 :3058	www.effetechn.com :80	/images/ed_small.gif	24269	FIN, 200
84	Ju...	192.168.1.3 :3061	www.etherdetect.com...	/images/logo_cornell.gif	2027	FIN, 200
85	Ju...	192.168.1.3 :3059	www.effetechn.com :80	/images/flag_detail.gif	1026	FIN, 200
86	Ju...	192.168.1.3 :3061	www.etherdetect.com...	/images/logo_reuters.gif	1822	FIN, 200
87	Ju...	192.168.1.3 :3059	www.effetechn.com :80	/images/flag_demo.gif	1013	FIN, 200
88	Tu...	192.168.1.3 :3059	www.etherdetect.com...	/images/flag_demo.gif	1018	FIN, 200

HTTP Request Header

```
GET /images/logo_ibm.gif HTTP/1.1
Accept: */*
Referer: http://www.etherdetect.com/
Accept-Language: zh-cn
Accept-Encoding: gzip, deflate
User-Agent: Mozilla/4.0 (compatible; MSIE 6.0; Windows NT 5.1)
Host: www.etherdetect.com
Connection: Keep-Alive
```

HTTP Response Header

```
HTTP/1.1 200 OK
Date: Sat, 07 Jun 2003 13:32:07 GMT
Server: Apache/1.3.27
Last-Modified: Mon, 14 Apr 2003 14:11:33 GMT
ETag: "bdae-4c1-3e9ac195"
Accept-Ranges: bytes
Content-Length: 1217
Keep-Alive: timeout=5, max=100
```

Ready Buffer: 3% URLs: 95 Packets: 393

Courtesy of EffeTech

HTTP Commands

HTTP Command	Description
GET	Request to read a webpage
HEAD	Request to read just the head section of a webpage
PUT	Request to write a webpage
POST	Request to append to a webpage
DELETE	Remove the webpage
LINK	Connects two existing resources
UNLINK	Breaks an existing connection between two resources

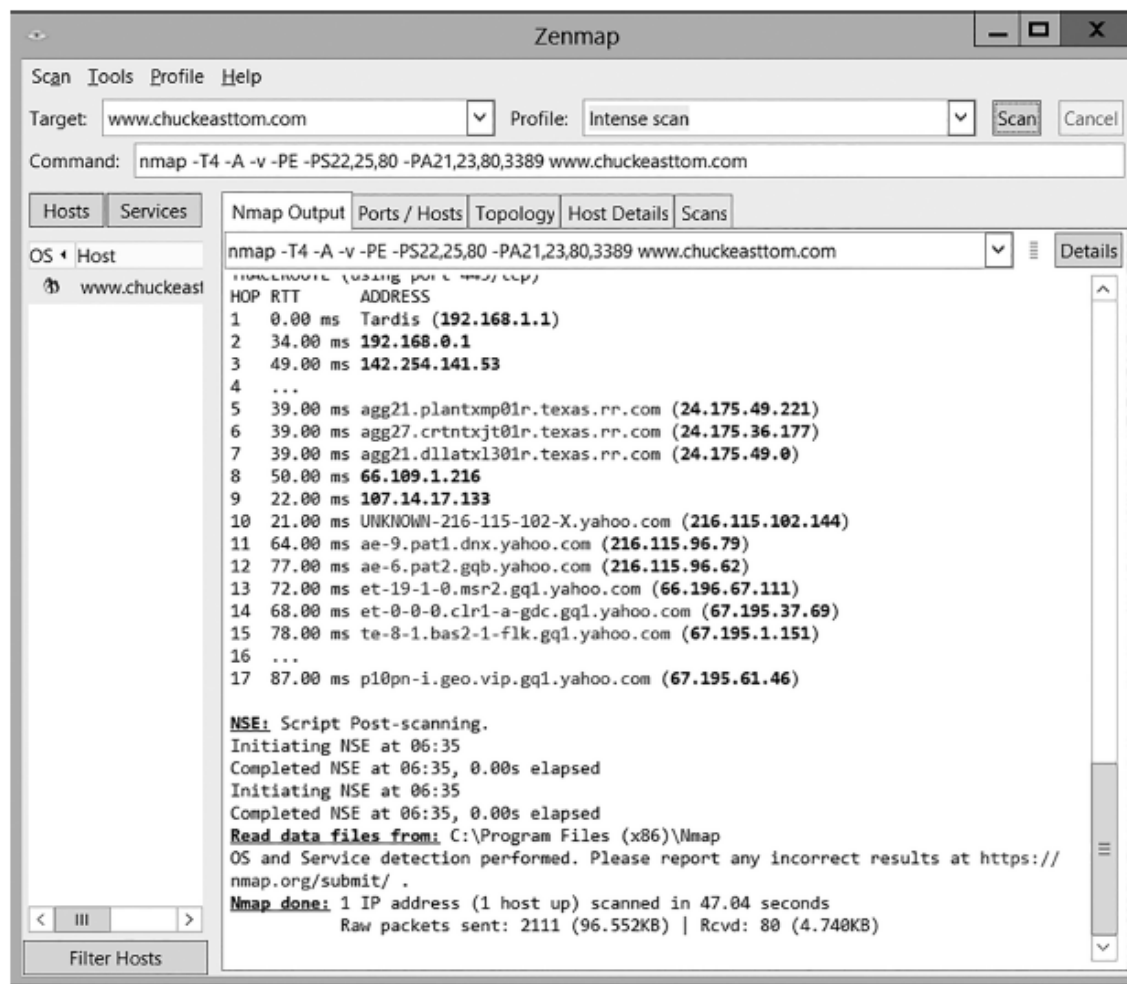
eTech

HTTP Response Messages

Message Range	Meaning
100–199	Informational; the server is giving your browser some information, most of which will never be displayed to the user
200–299	“OK” messages, meaning that whatever the browser requested, the server successfully processed
300–399	Redirect messages telling the browser to go to another URL
400–499	Client errors
500–599	Server-side errors

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Nmap/Zenmap



Courtesy of Nmap.org.

6.0.0.0

Snort

- Primarily used as an open source intrusion detection system
- Can function as a robust packet sniffer with a lot of configuration options

RSA NetWitness

- Threat analysis software/protocol analyzer
- Captures raw packets from wired and wireless interfaces
- Analyzes real-time data throughout the seven layers
- Filters by Media Access Control (MAC) address, Internet Protocol (IP) address, user, and more
- Freely available and threat analysis software

Network Traffic Analysis

- Can reveal:
 - Vulnerabilities and probing
 - Denial of service (DoS) attacks
 - User-to-root attacks
 - Remote-to-local attacks

Log Files as Source of Evidence

- Log files contain primary records of a person's activities on a system or network
- Log files can often identify:
 - Source, nature, and time of an attack
 - Specific user account of events related to illicit activities

Types of Logs

Authentication

Application

Operating system event

Network device

Security event

Challenges when Using Logs

- Logs change rapidly
- Getting permission to collect evidence from some sources, such as Internet service providers (ISPs), takes time
- Volatile evidence is easily lost
- Hackers can easily alter logs to include false information

Wireless Networking

Specification	Description
802.11a	• First widely used Wi-Fi standard • Operates at 5 GHz
802.11b	• Operates at 2.4 GHz • Indoor range of 125 ft • Bandwidth of 11 Mbps
802.11g	• Backward compatibility with 802.11b • Indoor range of 125 ft • Bandwidth of 54 Mbps

Wireless Networking (Cont.)

Specification	Description
802.11n	• Operates at 2.4 or 5.0 GHz • Indoor range of up to 230 ft • Bandwidth of 100 to 140 Mbps
IEEE 802.11n-2009	• Four spatial streams at a channel width of 40 MHz • Bandwidth of up to 600 Mbps • Uses multiple-input multiple-output MIMO

Wireless Networking (Cont.)

Specification	Description
IEEE 802.11ac	• Throughput of up to 1 Gbps with at least 500 Mbps • Uses up to eight MIMO
IEEE 802.11ad Wireless Gigabyte Alliance	• Data transmission rates up to 7 Gbps—more than 10 times faster than the highest 802.11n rate

Wireless Networking (Cont.)

■ Wardriving

- Process of driving around an area while a passenger in the vehicle scans for insecure, or weakly secured, wireless networks
- Participants then attempt to breach the targets they find

■ Warflying

- A twist on war driving
- A radio-controlled airplane is equipped with wardriving equipment
- Radio-controlled (RC) plane can be used to fly over an area to gather data on wireless networks

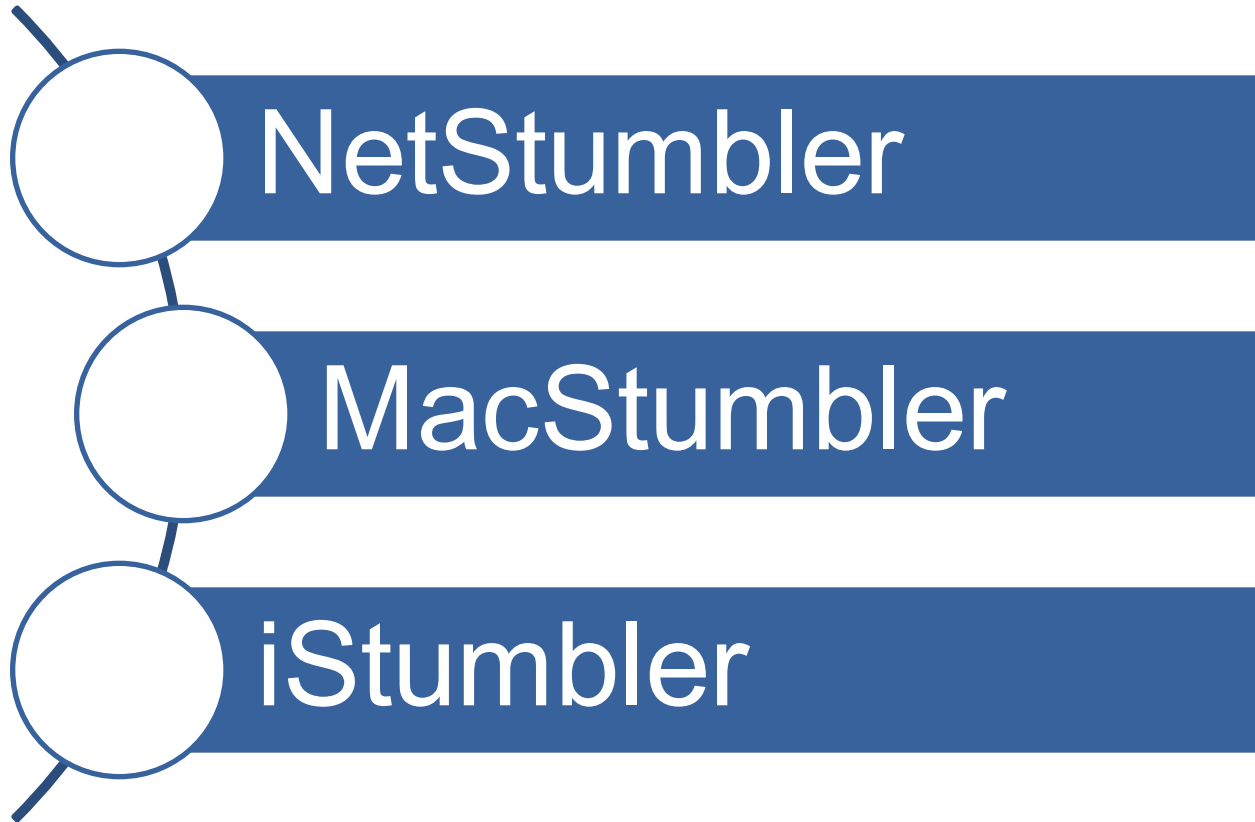
Wireless Concerns

- Did a perpetrator use a wireless network entry point for a direct network attack or theft of data?
- Did an attacker use a third-party wireless network, such as a hotel hotspot, to conceal his or her identity?

Wireless Storage Devices

- Wireless digital and video cameras
- Wireless printers with storage capacity
- Wireless network-attached storage (NAS) devices
- Tablets and smartphones
- Wireless digital video recorders (DVRs)
- Wireless game consoles

Wireless Network Discovery Tools



Router Forensics

- Router is hardware or software device that forwards data packets across a network to a destination network
- May contain:
 - Read-only memory (ROM) with power-on self test code
 - Flash memory containing the router's operating system
 - Nonvolatile random access memory (RAM) containing configuration information
 - Volatile RAM containing routing tables and log information

Basic Network Devices

Network card

- Enables computer to be connected to a network

Hub

- Connects computers on an Ethernet local area network (LAN)
- Sends any packet it receives out every port

Switch

- Prevents traffic jams by ensuring that data goes straight from its origin to its proper destination

Router

- Can connect different logical networks or subnets and enable traffic destined for the networks on the other side of the router to pass through

Routers in Detail

- Determine where to send information from one computer to another
- Are specialized computers that send your messages and those of every other Internet user to their destinations along thousands of pathways
- Maintain a routing table to keep track of routes
- Some routes are programmed manually, many are “learned” automatically by route

Router Attacks

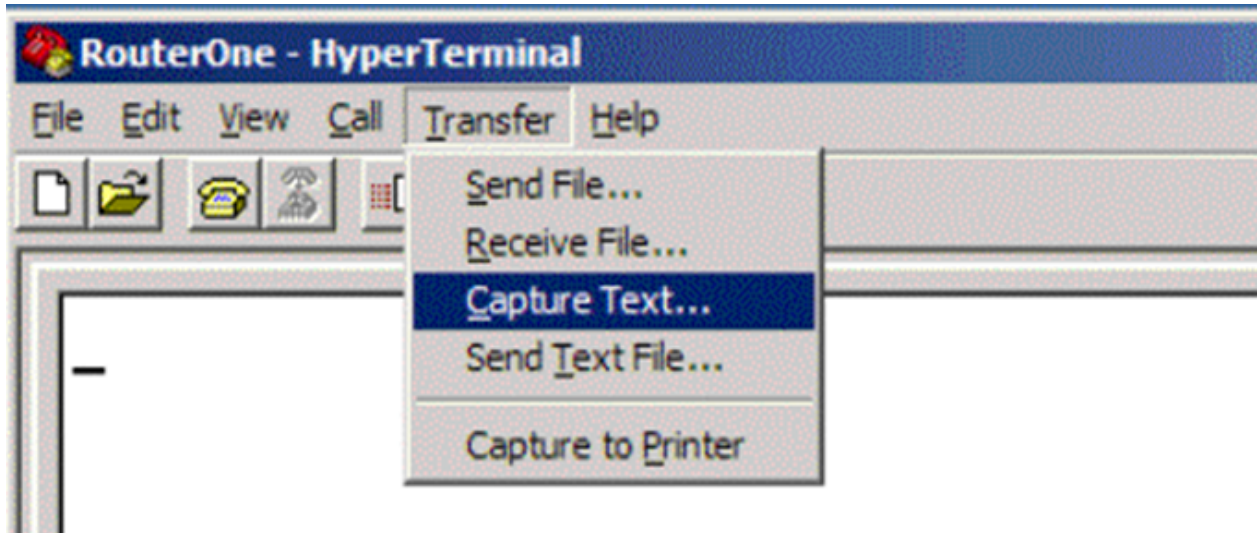
- Router table poisoning
- Incorrect router table entries can result in:
 - Artificial congestion
 - The router becoming overwhelmed
 - An attacker being allowed access to data in the compromised network

Getting Evidence from Routers

- Don't shut down router
 - Can erase valuable evidence
- Don't alter anything
- Document your process

- Connect with the router to run certain commands

Recording with HyperTerminal



Courtesy of HILGRAEVE

Router Commands

show
version

show
running-
config

show
startup-
config

show ip
route

show
interfaces

show
logging

What Is a Firewall?

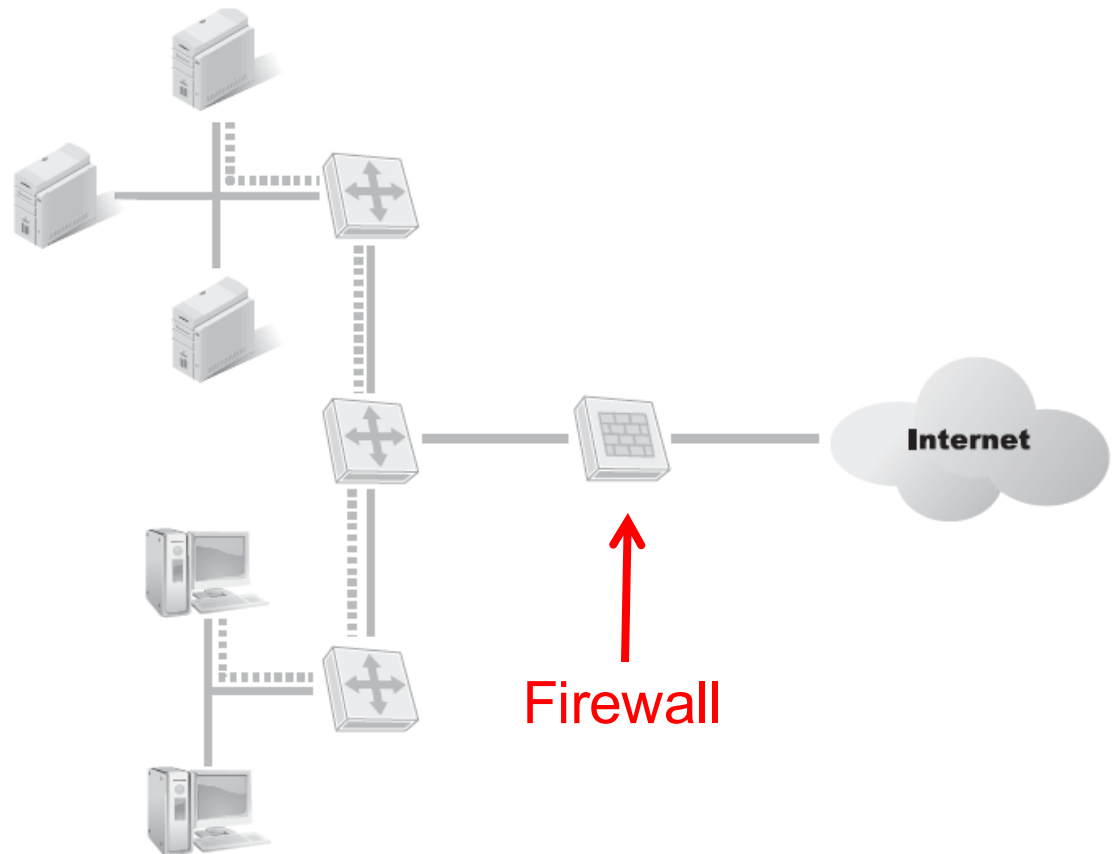
- A network traffic control device or service
- Enforces network security policy
- Protects the network against external attacks
- Establishes control over network traffic
- Prevents connections from unauthorized sources to protected network systems, services, and resources

Firewall Analogy

Bouncer at a night club with a guest list that defines specific names or types of individuals allowed in or specifically prohibited from the club

Network-Based Firewalls

- Span an entire network
- Filter all traffic passing in and out of network or network segment



Firewall Categories

Packet filter

Permits/denies incoming packets based on rules

Aka screened firewall

Stateful packet inspection

Examines all packets

Considers previous packets when permitting/denying

Collecting Data

All the traffic going through a firewall is part of a connection. A connection consists of two IP addresses communicating with each other and two port numbers that identify the protocol or service.

Type of Port	Port Numbers
Well-known ports	0 to 1023
Registered ports	1024 to 49151
Dynamic ports	49152 to 65535

Collecting Data (Cont.)

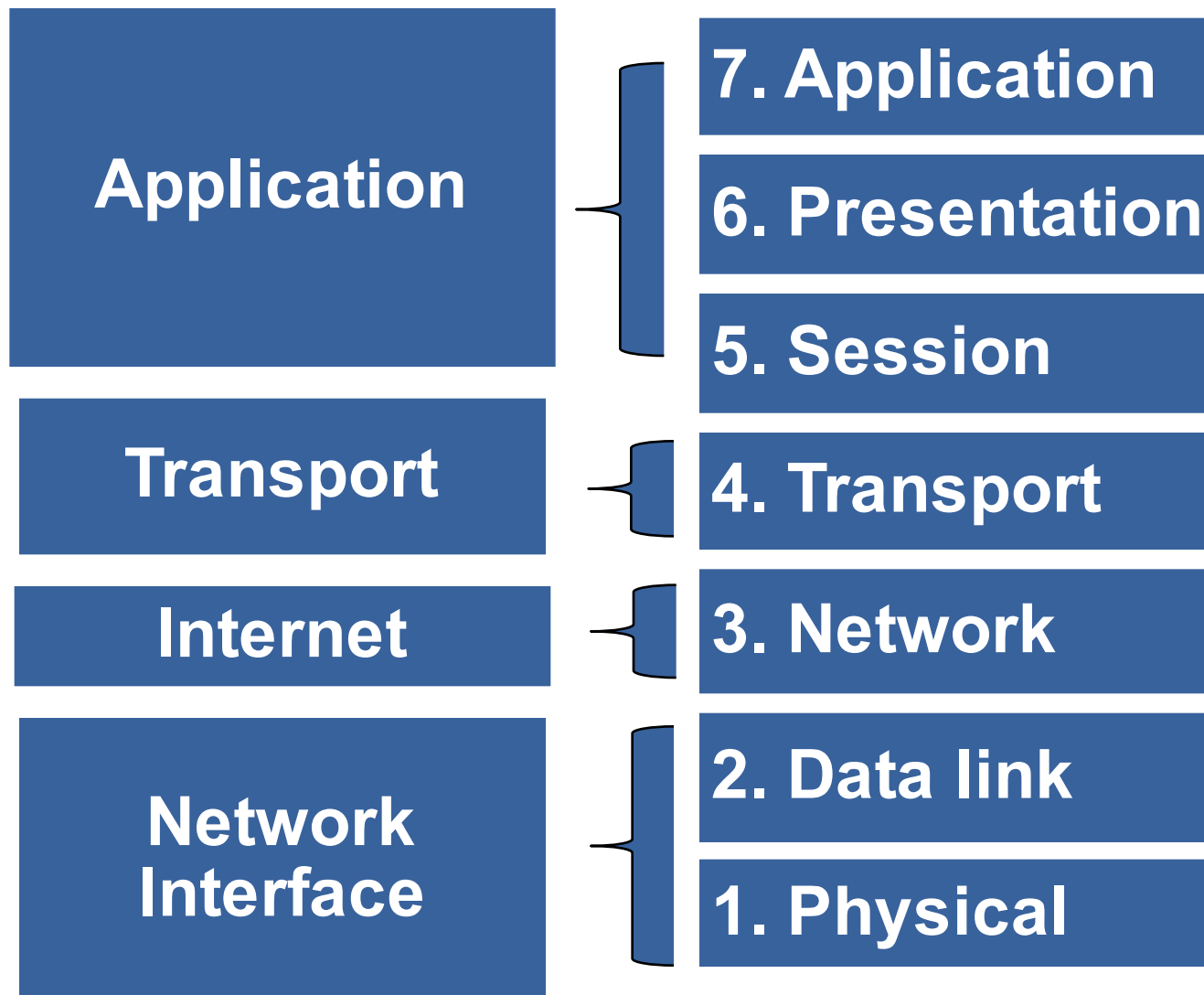
- Attempts on same set of ports from many different Internet sources are usually due to **decoy scans**
- Carefully check firewall logs for any sort of connections or attempted connections on those ports
- Use protocol analysis to determine who attacker is

Summary

- Network packet analysis basics
- Traffic analysis
- Router forensics
- Firewall forensics

OPTIONAL SLIDES

TCP/IP Networking and OSI Reference Models



TCP/IP Protocol Suite

Application

- Domain Name System (DNS), Dynamic Host Configuration Protocol (DHCP), Hypertext Transfer Protocol (HTTP), Tele-network (Telnet), File Transfer Protocol (FTP)

Transport

- Transmission Control Protocol (TCP), User Datagram Protocol (UDP)

Internet

- Internet Protocol (IP), IPsec, Internet Control Message Protocol (ICMP), Address Resolution Protocol (ARP), and Internet Group Management Protocol IGMP

Network Interface

- Serial Line Internet Protocol (SLIP), Purchasing Power Parity (PPP)

Why Analyze Data Packets?

- Detect network problems, such as bottlenecks
- Detect network intrusions
- Gather network statistics

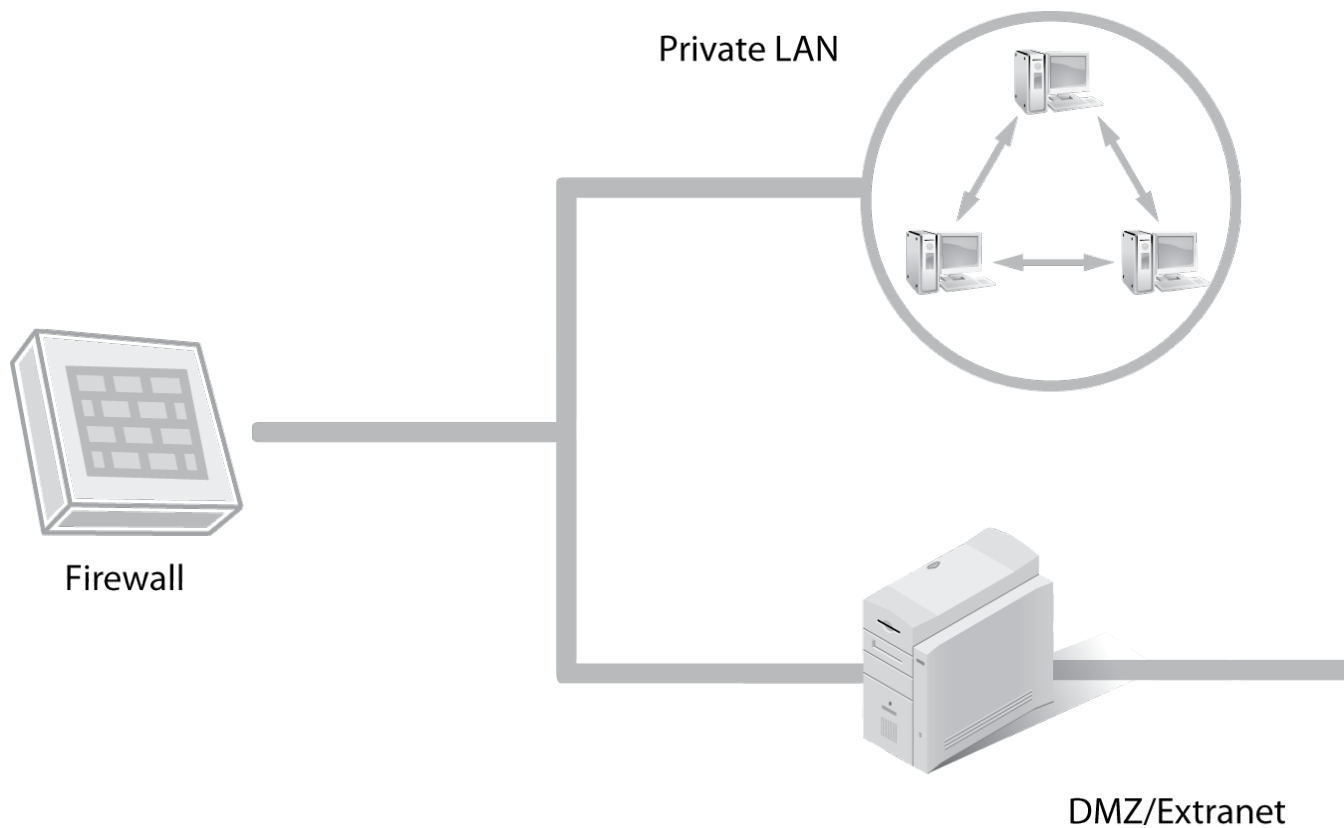
What Does a Protocol Analyzer Do?

- Captures and decodes data packets traveling on a network
- Allows you to read and analyze them

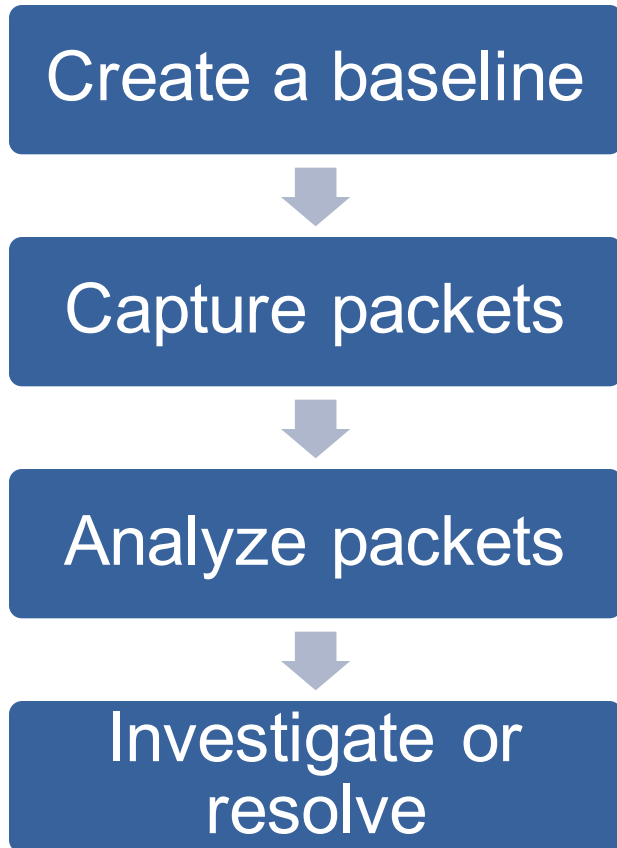
Where to Capture Data on the Network?

- Must first understand the network's architecture: where is the traffic of interest?
- Internal LAN usually generates too much traffic to analyze
- Monitor or capture at:
 - External (demilitarized zone, or DMZ) network
 - Just inside the perimeter firewall

Where to Capture Data on the Network? (Cont.)



Packet Capture and Analysis



MEASURE

Response time

Percentage of packets lost

TCP or UDP connection start and end

Examine Packets

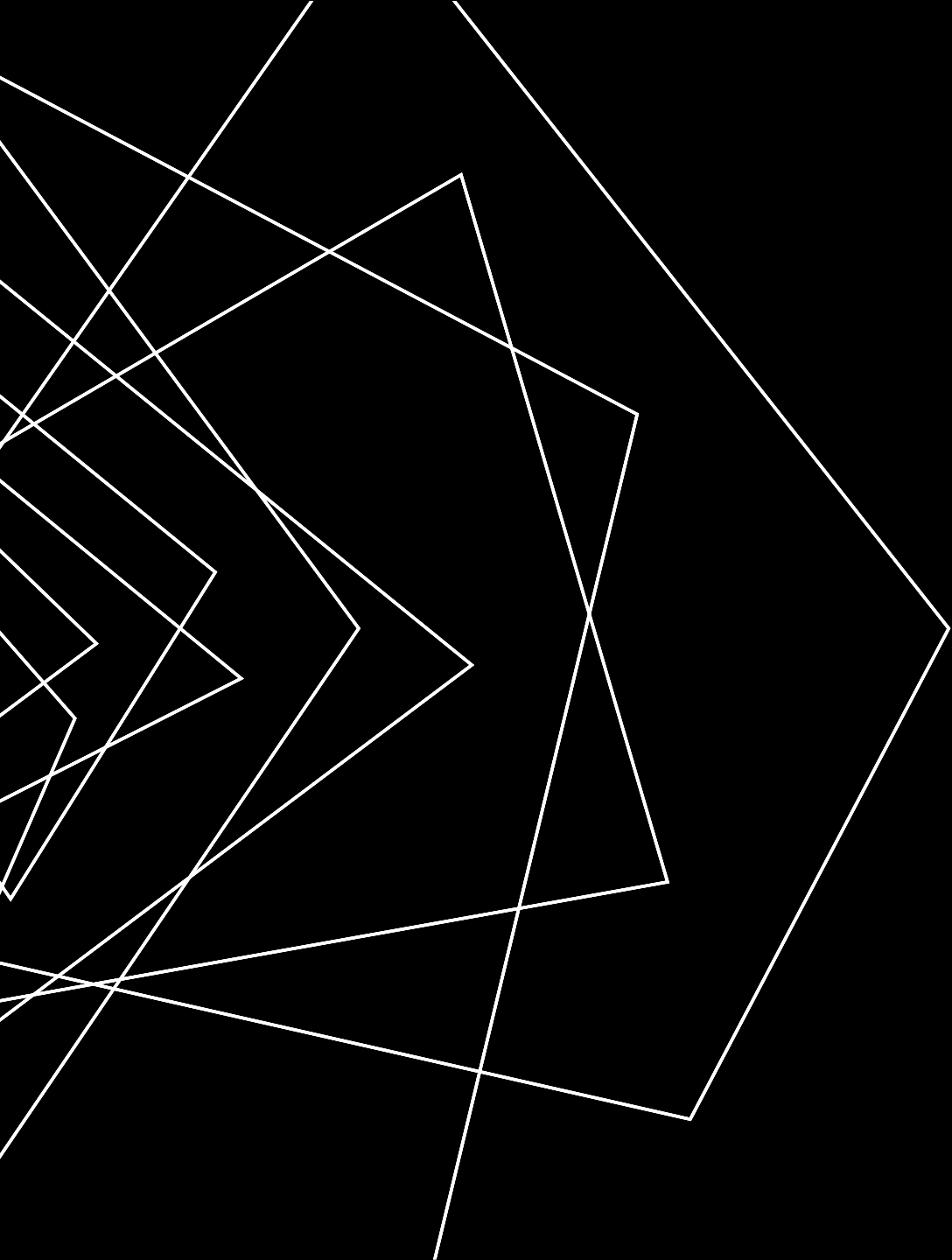
Normal packet

- Connecting to an FTP server
- Port 53 (dns) in User Datagram Protocol (UDP)
- Three-way handshake completes

Examine Packets (Cont.)

Packet showing evidence of port scan

- Series of TCP packets, part of three-way handshake
- Arrange segments in sequential order by source port
- Destination ports also in sequential order?
- Classic TCP port scan



THANK YOU

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